

ECONLOG POST
Feb 17 2026

TECHNOLOGY

Learning the Bitter Lesson in 2026

Joy Buchanan



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By Joy Buchanan, Feb 17 2026

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To prepare for teaching, I am reading a famous article in AI research: [The Bitter Lesson](#), written by Richard Sutton in 2019. I wondered what would seem prescient and if anything would feel like Sutton had gotten it wrong. At the end, I'll discuss economic implications.

Sutton draws from decades of AI history to argue that researchers have learned a “bitter” truth. Researchers repeatedly assume that computers will make the next advance in intelligence by relying on specialized human expertise. Recent history shows that methods that scale with computation outperform those reliant on human expertise. For example, in computer chess, brute-force search on specialized hardware triumphed over knowledge-based approaches. Sutton warns that researchers resist learning this lesson because building in

knowledge feels satisfying, but true breakthroughs come from computation's relentless scaling. In AI, scaling means making models larger and training them on more data with more compute.¹

The Bitter Lesson is less about any single algorithm than about intellectual humility: progress in AI has come from accepting that general-purpose learning, persistently scaled, outperforms our best attempts to hard-code intelligence. It matters whether Sutton is right or wrong, because we are not at the end of the explosion of AI or the period of time dubbed "[The Scaling Era](#)" by Dwarkesh Patel.

EconTalk guests have speculated that AI will save the world or kill us all. See the following:

- EconTalk: [Eliezer Yudkowsky on the Dangers of AI](#)
- EconTalk: [Erik Hoel on the Threat to Humanity from AI](#)
- EconTalk: [Marc Andreessen on Why AI Will Save the World](#)

Such extreme predictions assume that AI capabilities will advance. Although AI has been improving rapidly since Sutton wrote in 2019, there is no law of nature (that we know of) that insists it must continue to improve. Sometimes people even claim to see AI capabilities leveling off or point out that [hallucinations persist](#) even in advanced models.

If scaling is indeed the road to more intelligence, then we can expect AI to continue to exceed expectations if we add more hardware to the system. This hypothesis is being tested: US private AI investment could [exceed \\$100 billion annually](#), representing one of the largest technological bets ever. Let's examine Sutton's thesis in light of recent performance.

We can point to three pieces of evidence that Sutton was correct about scaling. First, game-playing AI provides a clean natural experiment. AlphaZero learned chess and Go through self-play, without human openings or strategy. AlphaZero surpassed earlier systems built on domain expertise. Its success came from scale and computation, just as Sutton predicted.

Second, natural language processing (NLP), the branch of AI focused on enabling computers to understand and generate human language, shows the same pattern. Earlier NLP systems emphasized linguistically informed rules and symbolic structure. OpenAI's GPT-3 and successors rely on generic architectures trained on vast data with enormous compute. Performance gains track scale more reliably than architectural cleverness.

The third example is computer vision. Hand-engineered feature pipelines (techniques where programmers manually designed algorithms to detect edges and shapes) were displaced once convolutional neural networks (a type of AI architecture loosely inspired by the visual cortex and designed to automatically learn visual patterns from data) could be trained at scale. Accuracy improved as datasets and compute increased.

Sutton's argument concerns the scalability of methods, but in practice that scalability only becomes visible once capital investment lowers computational constraints.

The rate of AI advancement reflects not just technological possibility but the unprecedented mobilization of financial resources. The typical person using ChatGPT to make grocery lists might not know what the word "scaling" means. A possible reason for underestimating the rate of progress is not just a misunderstanding of the technology but a missed estimate of how much money would be poured into it.

I compare this to the Manhattan Project. People doubted the Manhattan Project not because it violated physics, but because it seemed too expensive. Niels Bohr reportedly said it would require "turning the whole country into a factory." But we did it. We are doing it again. We are turning the country into a factory for AI. Without all that investment, the progress would be slower.

However, neither the doomers nor the utopians will turn out to be right if we are near a limit to either the power of scaling or our ability to physically continue to scale. Is the bitter lesson useful for seeing us through 2026 and beyond? This matters for unemployment today and existential threat tomorrow.

Recent economic research offers a nuanced view. In a January 2026 paper, economist Joshua Gans [develops a model](#) of “artificial jagged intelligence”. Gans observes that generative AI systems display uneven performance across tasks that appear “nearby”: they can be excellent on one prompt and confidently wrong on another with only small changes in wording or context. Anyone who has used ChatGPT to help with a work task and then watched it hallucinate a plausible-sounding falsehood has experienced this jaggedness firsthand.

What makes Gans’s analysis economically interesting is his treatment of scaling laws. In his model, increasing scale (represented by the density of known points in a knowledge landscape) shrinks average gaps and improves mean quality in a roughly linear fashion. This is good news for Sutton’s thesis: more compute does mean better average performance. However, jaggedness persists and errors remain. Scaling raises average performance without eliminating surprises or long-tail failures.

Gans frames AI adoption as an information problem: users care about local reliability (will the AI help *me* with *my* task?), but typically observe only coarse, global quality signals (benchmark scores). This mismatch creates real economic frictions. A legal assistant might trust an AI that performs brilliantly on 95% of contract reviews, only to be blindsided by a confidently wrong answer on a seemingly routine clause. The experienced errors, Gans shows, are amplified by what statisticians call the “inspection paradox”. Users encounter errors precisely in the gaps where they most need help.

Gans’s 2026 paper does not directly cite or refute Sutton, but it can be read as exploring a structural limitation that persists even when following the Bitter Lesson path. Scaling works, but the economic benefits of scaling may be partially offset by the persistent unpredictability that scaling does not cure.

This limitation has practical implications for how businesses adopt AI: they cannot simply trust benchmark performance but must invest in human oversight and domain-specific testing. This also means that AI will not spell the end of human jobs.

Sutton was right about the direction, but we shouldn't take his insight out of context. Scaling alone is not enough, and simply adding more scaling is unlikely to get us to superintelligence. Models still need human insight and structure to be maximally useful to companies. RLHF (Reinforcement Learning from Human Feedback), a training technique where human evaluators rate AI outputs to help the model learn which responses are helpful and safe, is an ingredient that injects human values into models. Earlier architectures didn't become GPT-4 only by adding more data.

Also, we cannot just "scale more" forever. Energy costs and data limits are real-world constraints. Thus, if AI is going to get much better it will need efficiency and algorithmic cleverness, not just brute force. Human insight has not faded into irrelevance yet. It has shifted from encoding intelligence directly to shaping, constraining, and steering scaled learning systems.

Overall, let's give Sutton due credit. Scaling works. But the efficiency of that scaling depends on human insight about how to structure and deploy these systems. Economists will recognize this as a familiar pattern: capital and labor remain complements, even when the capital is measured in GPUs and the labor involves designing loss functions.

Gans's work adds an important economic footnote: even as scaling improves average AI performance, the jagged, unpredictable nature of that performance creates real costs for adopters. Businesses and individuals must navigate a landscape where AI is simultaneously more capable and persistently unreliable in ways that are hard to anticipate. The economic returns to AI investment depend not just on raw capability but on developing institutions and complementary human expertise to manage jaggedness.

The bitter lesson may be that pure scaling is powerful, but the sweet corollary is that human ingenuity is still a vital ingredient for progress in the future.

[1] *Compute*, in AI research, is the total amount of computational power (typically measured in floating-point operations (FLOPs)) used to train or run a model.

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Brent Buckner

Feb 17 2026 at 8:04am



First, calling for better (near-universal) learning algorithms/architectures is in no way a refutation of the Bitter Lesson. Second, with the frontier AI labs driving toward automating AI research the direct human involvement in developing such algorithms/architectures may be much less than it seems that you're positing. Third, per Epoch AI, scaling at the recent trend through 2030 seems feasible, and that may produce Artificial General Intelligence at approximately the expert human level on remote-capable knowledge work.

Roger McKinney

Feb 25 2026 at 1:30pm



Very interesting! So far progress in AI has mostly been in language processing because it has trillions of documents to train on. The teal test of AI will be on businesses and economics where the data is sparse and unreliable.

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